Body Temperature Homeostasis



Thermoregulation **when body temperature rises**

Responses when body temperature rises above set point

- Vasodilation
  - Hypothalamus receives messages from thermoreceptors and initiates cooling response via autonomic NS.
  - Blood vessels in skin dilate
  - More blood can move through
  - Allows heat to be lost to surrounding environment

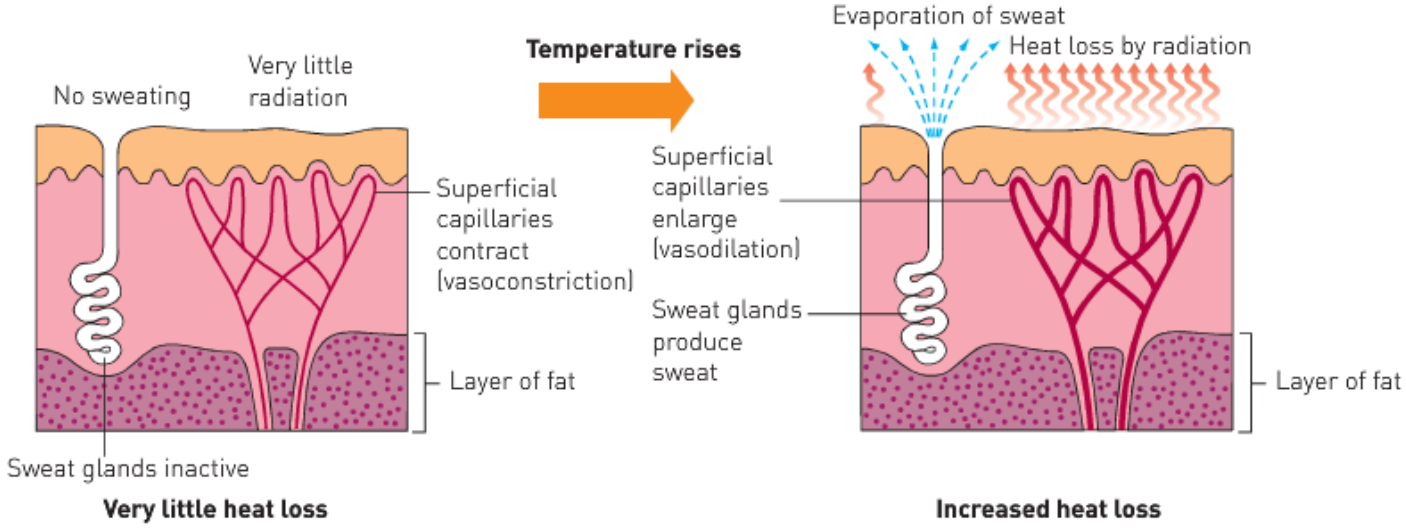


Figure 7.5 Heat loss from the skin can be increased by vasodilation of skin blood vessels and by sweating.

Learning Aim: Describe body responses when the body temperature rises above the set point

Responses when body temperature rises above set point

- Sweating
  - Hypothalamus receives messages from thermoreceptors and initiates cooling response via autonomic NS.
  - Needed for environmental temps above 28°C
  - Only option for cooling if environment above 37°C
  - Evaporation of liquid from skin accelerates heat transfer from body to environment
  - Works better in low humidity



Learning Aim: Describe body responses when the body temperature rises above the set point

Responses when body temperature rises above set point

- Conscious behavioural responses
  - Cerebral cortex receives information from thermoreceptors and initiates cooling response via the somatic motor NS (so we use our muscles to take action):
  - Wearing fewer clothing layers
  - Seeking cooler environments
  - Reducing physical activity



Learning Aim: Describe body responses when the body temperature rises above the set point

Responses when body temperature rises above set point

- Endocrine system – Thyroid
  - Rise in body temperature detected by thermoreceptors
  - Hypothalamus receives message and initiates response
  - Inhibits messages to thyroid (via the anterior pituitary) to inhibit thyroxine production
  - Less thyroxine means lower metabolic rate, so less heat produced.
  - This tends to be a much longer-term response: seasonal.



Learning Aim: Describe body responses when the body temperature rises above the set point

Homeostasis when Body Temp Rises

Stimulus
Rising body temperature

Receptor
Thermoreceptors in skin and hypothalamus

Message
Sent from skin and internal organs to hypothalamus via sensory NS. Hypothalamus also detects temp of blood.

Modulator (Control Centre)
Hypothalamus receives message that body temp is reising and activates cooling responses

Message
Sent via autonomic NS

Effector
Cerebral Cortex (conscious awareness)
Endocrine glands eg Thyroid
Blood Vessels in skin
Sweat Glands

Response
Body temp falls due to:
Conscious removal of clothing, shade seeking

Blood vessel dilation in skin - heat exchange
Sweat gland activity - evaporative cooling

Over time, metabolic rate may decrease - reduced Thyroxine

Feedback
Negative Feedback
Body temp returns to normal, stimulus ceases, cycle stops



Learning Aim: Draw detailed homeostatic feedback loops (steady-state control models) for thermoregulation when body temperature rises.

Homeostasis when body temperature rises



Stimulus

Rising Body Temperature

Receptor

Thermoreceptors in Skin (surface) and Hypothalamus, Spinal Cord and Abdominal Organs (core)

Message

Sensory Nervous System

Control Centre / Modulator

Hypothalamus

Cerebral Cortex Conscious Awareness)

Message

Autonomic Nervous System

Negative feedback

Somatic Motor NS

Effector

Blood Vessels

Sweat Glands

Thyroid

Skeletal Muscles

Response

Vasodilation allows greater blood flow to skin and heat exchange via convection and radiation

Sweat secretion allows evaporative cooling of skin surface.

Reduction in thyroid hormone production decreases metabolic rate and therefore heat production (longer term response)

Behavioural responses such as wearing fewer clothing layers, seeking shade/cool environment, reducing physical activity.

Feedback

Negative Feedback: Body temperature returns to normal, stimulus no longer present, process ends.

Learning Aim: Draw detailed homeostatic feedback loops (steady-state control models) for thermoregulation when body temperature rises.

In cold conditions

To decrease heat loss

- Constriction of blood vessels in the skin
- Reduction in sweating
- Conscious behaviour such as putting on a jumper, sheltering from wind
- Reduction of surface area such as by curling into a ball



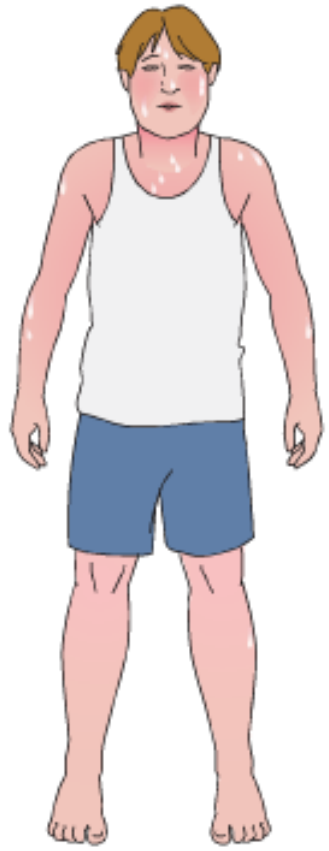
To increase heat production

- Shivering
- Increase in voluntary activity
- Increased metabolic rate (long-term response)

In hot conditions

To increase heat loss

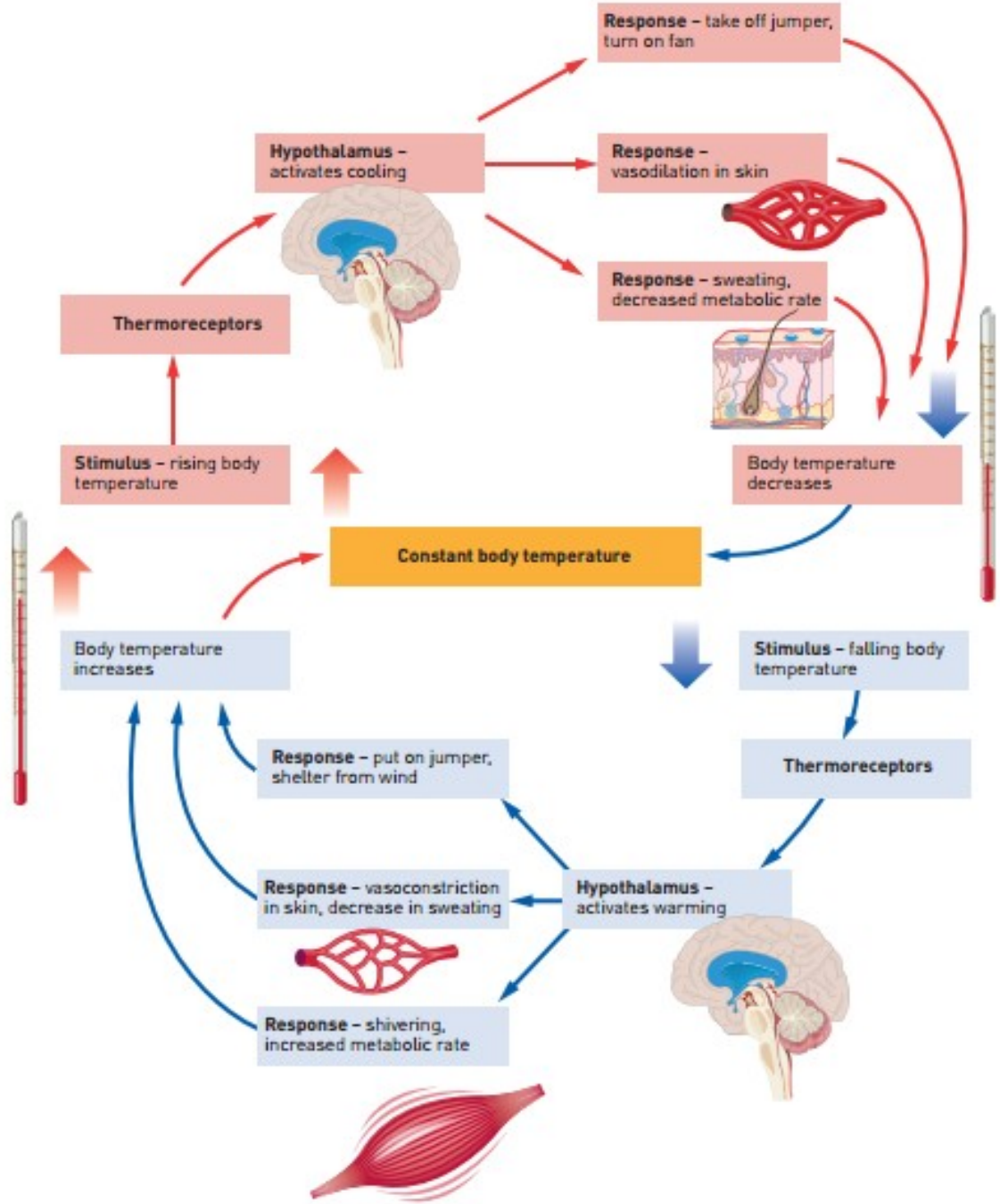
- Dilation of blood vessels in the skin
- Sweating
- Conscious behaviour such as removing clothing, turning on a fan
- Increase surface area by spreading out



To decrease heat production

- Decrease in voluntary activity
- Decreased metabolic rate (long-term response)

Figure 7.6 Mechanisms for regulating temperature


Response – take off jumper, turn on fan
Hypothalamus – activates cooling
Response – vasodilation in skin
Response – sweating, decreased metabolic rate
Thermoreceptors
Stimulus – rising body temperature
Body temperature decreases
Constant body temperature
Body temperature increases
Stimulus – falling body temperature
Response – put on jumper, shelter from wind
Thermoreceptors
Response – vasoconstriction in skin, decrease in sweating
Hypothalamus – activates warming
Response – shivering, increased metabolic rate

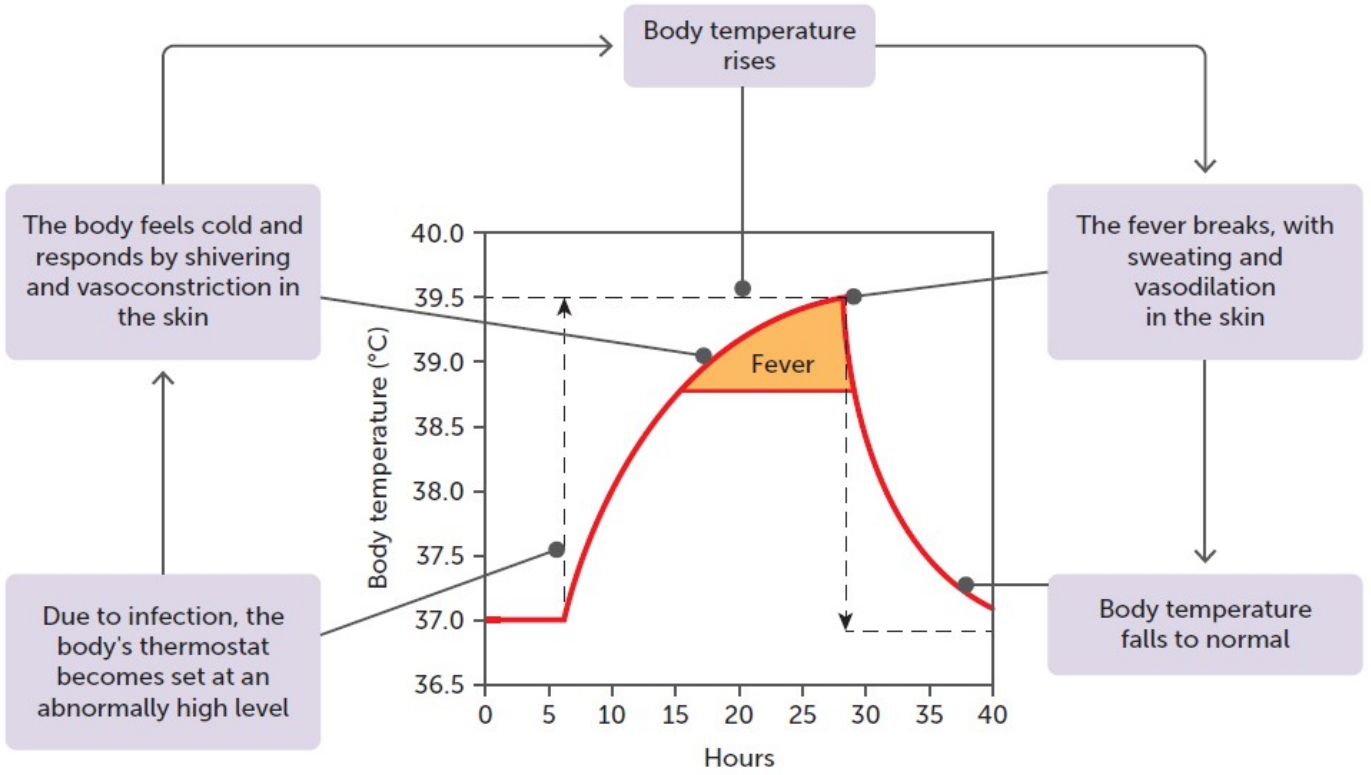
Consequences when tolerance limits are exceeded.

- Core body temp above 42°C is dangerous and will result in death if prolonged.
- Heat stroke:
  - when the body is unable to manage temperature due to extreme external conditions, and core body temperature rises above tolerance limits.
  - Treatment – careful cooling.
- Heat exhaustion:
  - Less severe
  - Dehydration from sweating + vasodilation → drop in BP and fainting.
  - Core body temperature may not change – it's a drop in BP because of the normal cooling responses.
  - Treatment – help patient to feel cooler (cold compress, cool drink) and give fluids.
- Hypothermia:
  - If body temperature drops below 32°C.
  - Death if prolonged
  - Treatment – careful, slow warming.

Learning Aim: Discuss consequences to the body when tolerance limits for body temperature are breached.

Fever – a response that raises core body temperature

- Part of the body's natural immune response to infection.
- Body raises temperature to combat pathogens such as viruses or bacteria.
- After encountering infection, white blood cells release *pyrogens* (fever causing chemicals) eg: interleukin-1
- Pyrogens affect the hypothalamus so that the normal set point of 37°C is temporarily reset to a higher temperature.
- Hypothalamus then initiates warming responses to raise body temperature to the new, temporary set point.
- These include normal warming responses such as shivering, vasoconstriction, seeking warmth etc, until the body temperature reaches the new "set point".
- Once the infection has been dealt with, the immune system stops producing pyrogens, and the "set point" returns to normal.
- The person then feels hot and cooling responses are activated eg sweating and vasodilation.
- Fever is beneficial, but only up to a point. Prolonged, high fever can cause harm and even death, if the immune system is unable deal with the infection quickly and effectively.



Learning Aim: Discuss why fever occurs and the physiological processes involved.